2 Describe the format of a database table.

3 Identify the purpose of a primary key.

4 a A social media website stores the following information about its users: username (e.g. daisy123), password (e.g. HWHRU2D), online name (e.g. Daisy), date account created (e.g. 9/9/2019), number of friends (e.g. 38).
Identify the fields and data types for a single-table database for the website.

b Identify an appropriate primary key for the table.

# 9.2 SQL

**SQL** stands for Structured Query Language. This is a standard language that is used across most types or database to perform actions such as:

- define tables
- change tables
- add data to tables
- search for data from tables (query)
- perform calculations using data from tables.

In this specification you only need to know the last two of these: search for data and perform calculations using data.

When you write some SQL code you are writing an SQL **script.** This is series of statements that are commands that are then executed and one or more values are returned.

KEY WORDS

**Structured Query Language (SQL):** a standard language used to define and manipulate databases.

**script:** a set of statements that are executed.

# 9.3 SELECT ... FROM

This allows you select a set of fields from a table. All of the data in each field will be returned.

This table is called Books.

| Book ID | BookName | Author | Publisher | Genre | Fiction |
|---|---|---|---|---|---|
| 1 | Picking daisies | J. Frank | Cambridge horticulture | Gardening | False |
| 2 | Night stars | K. Mars | Si-fi books | Science fiction | True |
| 3 | Dreaming of the sun | P. Yu | Si-fi books | Science fiction | True |
| 4 | Cooking for fun | W. Crisp | Cookery penguin | Cookery | False |

Select has the format:

```
SELECT field
FROM table
```

Single field:

```
SELECT BookName
FROM Books
```

This query will return:

Picking daisies
Night stars
Dreaming of the sun
Cooking for fun

**IMPORTANT:** The field names and table name must be exact. If the field name in the table is 'Genre' then putting 'genre' in the query is incorrect. It is case sensitive.

Multiple fields:

```
SELECT BookName, Genre
FROM Books
```

Notice how there are no commas or any other symbol between each field.

This will return

| | |
|---|---|
| Picking daisies | Gardening |
| Night stars | Science fiction |
| Dreaming of the sun | Science fiction |
| Cooking for fun | Cookery |

The order the fields appear in the query are important. The data will be returned in the order they appear in the query.

Example:

```
SELECT BookName, Author
FROM Books
```

will return book name then author:

| | |
|---|---|
| Picking daisies | J. Frank |
| Night stars | K. Mars |
| Dreaming of the sun | P. Yu |
| Cooking for fun | W. Crisp |

The command SELECT is followed by the field name.

The command FROM is followed by the name of the table.

The case, spellings and order are all important in the answer. Putting commas after each one is incorrect. Putting them one after another on a line is incorrect. Each record is on a new line.

Each field is separated by a comma (,).

Each record is on a new line, Picking daisies and Gardening are on the same line because they are part of the same record.

whereas:

```
SELECT Author, BookName
FROM Books
```

will return author then book name:

| | |
|---|---|
| J. Frank | Picking daisies |
| K. Mars | Night stars |
| P. Yu | Dreaming of the sun |
| W. Crisp | Cooking for fun |

**COMPUTER SCIENCE IN CONTEXT**

SQL is known as a standard. This means that it is used (almost) universally, no matter what software you are using, or type of database. This means that anyone who knows SQL can use any type of database simply by entering the same SQL code.

## Questions

These questions use the following database table, PRODUCTS, about products for sale in a shop.

| ID number | Name | Type | Cost | Quantity in stock |
|---|---|---|---|---|
| 12GH | Saturn | Chocolate | 0.56 | 100 |
| 59RF | Vanilla | Fudge | 1.99 | 20 |
| 3WR | Peanuts | Nuts | 2.56 | 2 |
| 52D | Jazz | Fruit | 0.23 | 33 |
| 6SE | Signet | Fudge | 1.48 | 5 |

**5** Write the results returned by the following SQL script:

```
SELECT ID number
FROM PRODUCTS
```

**6** Write the results returned by the following SQL script:

```
SELECT Name, Type
FROM PRODUCTS
```

**7** Complete the SQL script to display the prices of all the products.

```
SELECT .....
FROM PRODUCTS
```

**8** Complete the SQL script to display the ID number and name of all the products.

```
..... ID Number, .....
FROM PRODUCTS
```

9 Complete the SQL scripts to display the Name, Cost and Quantity in stock for all items.

```
..... ....., ....., Quantity in stock
..... PRODUCTS
```

# 9.4 SELECT ... FROM ... WHERE

Including a WHERE clause in the SELECT statement allows you to select only specific data. For example, you only want gardening books, or items that cost more than $3.00.

## Single condition (single WHERE)

This has only one condition. The condition needs a **logical operator**. These are the same operators that you use in programming (Table 9.2).

KEY WORD

**logical operator:** a symbol that performs a comparison resulting in True or False. Can be equals, not equal to, less than, less than or equal to, greater than, greater than or equal to.

| Operator | Description | Example WHERE |
|---|---|---|
| = | Equal to. | WHERE Fiction = True |
| < | Less than. | WHERE Cost < 3.99 |
| <= | Less than or equal to. | WHERE Cost <= 10 |
| > | Greater than. | WHERE Quantity > 0 |
| >= | Greater than and equal to. | WHERE Quantity >= 100 |
| <> | Not equal to. | WHERE Cost <> 4.0 |

**Table 9.2:** Logical operators

The following queries will use the following table, `Orders`:

| Order ID | FirstName | LastName | NumberItems | TotalCost | Posted |
|---|---|---|---|---|---|
| 1A | Qirat | Raja | 20 | 21.99 | No |
| 1B | Simon | Tong | 2 | 3.99 | Yes |
| 1C | Mei | Liu | 1 | 6.00 | Yes |
| 1D | Jenny | Silver | 29 | 59.65 | No |

Select where has the format:

```
SELECT SUM field
FROM table
WHERE condition
```

**Equal to**

```
SELECT FirstName, LastName, Posted
FROM Orders
WHERE Posted = Yes
```

This will return:

| | | |
|---|---|---|
| Simon | Tong | Yes |
| Mei | Liu | Yes |

**Not equal to**

```
SELECT Order ID, Posted
FROM Orders
WHERE Posted <> Yes
```

This query returns:

| | |
|---|---|
| 1A | No |
| 1D | No |

**Less than**

```
SELECT Order ID
FROM Orders
WHERE NumberItems < 10
```

This query returns:

1B
1C

**Less than or equal to**

```
SELECT NumberItems, TotalCost
FROM Orders
WHERE TotalCost <= 6.00
```

This query returns:

| | |
|---|---|
| 2 | 3.99 |
| 1 | 6.00 |

**Greater than**

```
SELECT FirstName, LastName
FROM Orders
WHERE NumberItems > 20
```

This query returns:

| | |
|---|---|
| Jenny | Silver |

**Greater than or equal to**

```
SELECT Posted, FirstName
WHERE TotalCost >= 3.99
```

This query returns:

| | |
|---|---|
| Qirat | No |
| Simon | Yes |
| Mei | Yes |
| Jenny | No |

## Questions

These questions use the following database table, MARKS, about students and their marks in tests.

| Student ID | Subject | Test name | Percentage | Grade |
|---|---|---|---|---|
| 123 | Science | Physics 1 | 10 | Fail |
| 596 | Computer Science | Programming 2 | 53.3 | Pass |
| 123 | Maths | Trigonometry 6 | 10 | Pass |
| 023 | Maths | Trigonometry 6 | 50 | Pass |
| 802 | Science | Physics 1 | 25 | Fail |
| 806 | Computer Science | Programming 2 | 46.6 | Pass |
| 023 | Maths | Trigonometry 7 | 81 | Merit |
| 596 | Science | Physics 1 | 50 | Pass |

**10** Write the results returned by the following SQL script:

```
SELECT Student ID, Subject
FROM MARKS
WHERE Subject = "Programming 2"
```

**11** Write the results returned by the following SQL script:

```
SELECT Student ID, Subject, Grade
FROM MARKS
WHERE Grade = "Pass"
```

**12** Complete the SQL script to display the Test name, percentage and grade where the percentage is more than 50.

```
..... Test name, Percentage, .....
FROM .....
WHERE Percentage ..... .....
```

**13** Complete the SQL script to display the student ID, subject and percentage where the subject is Science.

```
SELECT Student ID, ....., .....
..... MARKS
..... SUBJECT = .....
```

**14** Complete the SQL script to display the student ID, test name and percentage where the percentage is less than or equal to 25.

```
..... ....., Test name, Percentage
..... .....
WHERE ..... <= .....
```

## More than one condition

Combining conditions requires a **Boolean operator** between the conditions. These are similar to those that you used in programming (Table 9.3).

KEY WORD

**Boolean operator:** a symbol that joins multiple logical comparisons, can be AND OR or NOT.

| Operator | Description | Example WHERE |
|---|---|---|
| AND | Only data that meets both conditions will be returned. | WHERE Fiction = TRUE AND Cost < 3.99.<br><br>Only records where Fiction is TRUE and Cost is less than 3.99 will be displayed. |
| OR | Data that meets one condition, or both conditions will be returned. | WHERE Fiction = TRUE OR Cost < 3.99.<br><br>Any records where Fiction is TRUE will be returned.<br>Any records where Cost < 3.99 will be returned.<br><br>If a record meets both then this will be returned only once. |

**Table 9.3:** Boolean operators

Multiple conditions have the format:

```
SELECT field
FROM table
WHERE condition1 logical operator condition2
```

The following queries will use the following table, `Orders`:

| Order ID | FirstName | LastName | NumberItems | TotalCost | Posted |
|---|---|---|---|---|---|
| 1A | Qirat | Raja | 20 | 21.99 | No |
| 1B | Simon | Tong | 2 | 3.99 | Yes |
| 1C | Mei | Liu | 1 | 6.00 | Yes |
| 1D | Jenny | Silver | 29 | 59.65 | No |

**AND**

```
SELECT Order ID, NumberItems, Posted
FROM Orders
WHERE NumberItems > 1 AND Posted = Yes
```

This query returns:

| | | |
|---|---|---|
| 1B | 2 | Yes |

**OR**

```
SELECT FirstName, TotalCost, Posted
FROM Orders
WHERE TotalCost < 5 OR Posted = No
```

This query returns:

| | | |
|---|---|---|
| Qirat | 21.99 | No |
| Simon | 3.99 | Yes |
| Jenny | 59.65 | No |

## Selecting values in, or not in, a condition

The fields in the SELECT do not need to include those in the WHERE.

Example:

```
SELECT FirstName, LastName
FROM Orders
WHERE Posted = Yes
```

This will return the first name and last name of orders where posted is true. It does not return the field posted. It will return:

| | |
|---|---|
| Simon | Tong |
| Mei | Liu |

### COMPUTER SCIENCE IN CONTEXT

If you log into a website then the code behind the scenes runs a query to find out what your username and password are, and then compares what you entered to these details. This is likely to use an SQL SELECT ... FROM ... WHERE statement.

## Questions

These questions use the following database table, `WEATHER`, that store weather readings.

| Date | Time | Temperature | Wind speed | Humidity |
|---|---|---|---|---|
| 11/12/2019 | 10:00 | 20.3 | 22 | 59 |
| 11/12/2019 | 13:00 | 23.7 | 15 | 63 |
| 11/12/2019 | 15:00 | 24.0 | 13 | 61 |
| 11/12/2019 | 18:00 | 23.9 | 16 | 60 |
| 12/12/2019 | 15:00 | 18.4 | 20 | 52 |
| 12/12/2019 | 18:00 | 17.2 | 18 | 48 |

**15** Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
WHERE Date = 11/12/2019 AND time < 15:00
```

**16** Write the results returned by the following SQL script:

```
SELECT Date, Time, Humidity
FROM WEATHER
WHERE Temperature > 20 AND Wind speed < 20
```

**17** Write the results returned by the following SQL script:

```
SELECT Date, Time
FROM WEATHER
WHERE Temperature > 20 OR Wind speed >= 20
```

**18** Complete the SQL script to display the time when the date is 12/12/2019 and the humidity is more than 50.

```
SELECT .....

FROM .....

WHERE .....= 12/12/2019 ..... Humidity > 50
```

**19** Complete the SQL script to wind speed and temperature, when the wind speed is either below 15 or above 20.

```
..... Wind speed, .....

FROM WEATHER

WHERE Wind speed ..... 15 ..... Wind speed ..... 20
```

## PROGRAMMING TASK 9.2

These questions will need your shop database table from Programming Task 9.1.

### Getting started

**1** Identify the command words in an SQL search script.

**2** Identify *four* logical operators and describe the function of each.

**3** Identify *two* Boolean operators and describe the function of each.

### Practice

**1a** Complete the SQL script to return the name of all the products in the store.

```
SELECT .....

FROM .....
```

**1b** Identify the values that will be returned from your query in **1a**.

**2a** Complete the SQL script to return the name of all products where an order for more products have been made.

```
SELECT .....

FROM .....

WHERE ..... ..... .....
```

**2b** Identify the values that will be returned from your query in 2a.

**3a** Complete the SQL script to return the product barcode, colour and size where the quantity in stock is less than 10 and an order for more products has not been made.

```
SELECT .....,  .....,  .....

FROM .....

WHERE ..... ..... .....

AND ..... ..... .....
```

**3b** Identify the values that will be returned from your query in **3a**.

**CONTINUED**

**Challenge**

This task goes beyond the range of the specification.

Find out how to write SQL queries in the database software you have used. Write the queries for practice and run them in your database.

# 9.5 Order by

This command allows you to state which order the result will be displayed in. There are two options, ascending (`ASC`) and descending (`DESC`). The default is ascending.

The order by statement comes after the `SELECT ... FROM` or `SELECT ... FROM ... WHERE`.

The following queries will use the following table, `Orders`:

| Order ID | FirstName | LastName | NumberItems | TotalCost | Posted |
|---|---|---|---|---|---|
| 1A | Qirat | Raja | 20 | 21.99 | No |
| 1B | Simon | Tong | 2 | 3.99 | Yes |
| 1C | Mei | Liu | 1 | 6.00 | Yes |
| 1D | Jenny | Silver | 29 | 59.65 | No |

## Example 1:

```
SELECT OrderID, NumberItems
FROM Orders
ORDER BY NumberItems ASC
```

This query will return:

| | |
|---|---|
| 1C | 1 |
| 1B | 2 |
| 1A | 20 |
| 1D | 29 |

## Example 2:

```
SELECT OrderID, TotalCost, Posted
FROM Orders
WHERE Posted = Yes
ORDER BY TotalCost DESC
```

This query will return:

| | | |
|---|---|---|
| 1C | 6.00 | Yes |
| 1B | 3.99 | Yes |

## Questions

These questions use the following database table, WEATHER, that store weather readings.

| Date | Time | Temperature | Wind speed | Humidity |
|---|---|---|---|---|
| 11/12/2019 | 10:00 | 20.3 | 22 | 59 |
| 11/12/2019 | 13:00 | 23.7 | 15 | 63 |
| 11/12/2019 | 15:00 | 24.0 | 13 | 61 |
| 11/12/2019 | 18:00 | 23.9 | 16 | 60 |
| 12/12/2019 | 15:00 | 18.4 | 20 | 52 |
| 12/12/2019 | 18:00 | 17.2 | 18 | 48 |

**20** Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
ORDER BY Temperature
```

**21** Write the results returned by the following SQL script:

```
SELECT Date, Time, Temperature
FROM WEATHER
ORDER BY Humidity DESC
```

**22** Complete the SQL script to display the date, wind speed and humidity in ascending order of wind speed.

```
..... Date, Wind speed, Humidity
..... WEATHER
..... Wind speed
```

**23** Complete the SQL script to display the date, time and temperature, in descending order of humidity.

```
SELECT Date, Time, .....
FROM WEATHER
ORDER BY ..... .....
```

**24** Complete the SQL script to display the temperature, wind speed and humidity in descending order of temperature.

```
..... ....., Wind speed, .....
FROM .....
..... ..... .....
```

# 9.6 SUM

SUM command will add up the field that is written after it. It is written in the select statement in the format:

```
SELECT SUM(field)
FROM table
```